

Exercises

1	2	3	4	5	6	7	8	9	10
11									

Surname, First name

Statistics (BCS1520)

BCS1520 Mock Exam 2024-2025

1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
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☐ a ☒ b ☐ c ☐ d ☐ e ☐ f → b
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☐ a ☒ b ☐ c ☒ d ☐ e ☐ f → b, d
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☒ a ☐ b ☒ c ☒ d ☐ e ☐ f → c

Answer multiple-choice questions as shown in the example. Circular checkboxes have only one correct answer. Square checkboxes may have multiple correct answers.

Program: Bachelor Computer Science

Course code: BCS1520

Examiners: Anirudh Wodeyar, Tim Dick, Niloufar Yousefimanesh, and Luuk Verkleij

Date/time: 23-5-2025 9:00h-11:00h

Format: Closed book

Allowed aids: Pen, and simple calculator from DACS approved calculator list, formula sheet (provided)

Instructions to students:

- The exam consists of 10 questions (plus question 11 for extra space for answers) on 22 pages.
- Fill in your name and student ID number on the cover page and tick the corresponding numerals of your student number in the table (top right cover page).
- Answer every question in the reserved space below the question. **Do not write outside the reserved space or on the back of pages, this will not be scanned and will NOT be graded!** As a last resort if you run out of space, use the extra answer space at the end of the exam.
- *In no circumstance write on or near the QR code at the bottom of the page!*
- Ensure that you properly motivate your answers.
- Only use black or dark blue pens, and write in a readable way. Do not use pencils.
- Answers that cannot be read easily cannot be graded and may therefore lower your grade.
- If you think a question is ambiguous, or even erroneous, and you cannot ask during the exam to clarify this, explain this in detail in the space reserved for the answer to the question.
- If you have not registered for the exam, your answers will not be graded, and thus handled as invalid.
- You are not allowed to have a communication device within your reach, nor to wear or use a watch.
- You have to return all pages of the exam. You are not allowed to take any sheets, even blank, home.
- The points in this exam sum up to 110.
- Good luck!

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Multiple Choice Questions

2p **1a** When you perform a **double-blind** experiment, who has knowledge whether a subject in the study is in the treatment or control condition?

- (a) Neither the subjects, nor the doctors know but the statisticians do.
- (b) The subjects and statisticians do know, but the doctors do not know.
- (c) The subjects, the doctors and statisticians don't know.
- (d) The doctors and statisticians do know, but the subjects do not know.

3p **1b** In a certain university, the admissions committee released the following summary for two departments. (Numbers are **admitted / applicants**.)

Department	Women	Men
A	5 / 5	45 / 50
B	35 / 95	3 / 15
Total (both depts.)	40 / 100	48 / 65

Which ONE of the following statements is **correct**?

- (a) Women have the higher admission rate in each department, yet men have the higher admission rate overall.
- (b) Women have a higher admission rate than men overall, but men have the higher rate in Department B.
- (c) Men have the higher admission rate in each department and overall.
- (d) Women have a higher admission rate than men in each department *and* overall.

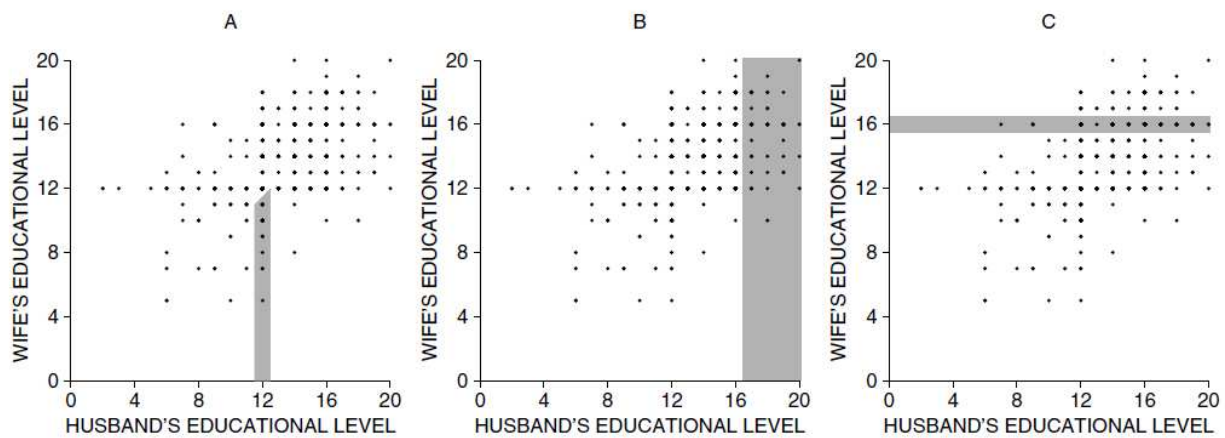
- 3p **1c** The table of test scores below has an average of 50 and a standard deviation (SD) of 10.

39	41	47	58	65	37	37	49	56	59	62	36	48
52	64	29	44	47	49	52	53	54	72	50	50	

Assume this data comes from a bell curve and use that to determine how many scores are 1.25 SDs or more away from the average. The total number of observations that is more than 1.25 SDs away of the average is:

- ☐ a 10 observations
- ☐ b 6 observations
- ☐ c 7 observations
- ☐ d 9 observations
- ☐ e 8 observations
- ☐ f 5 observations

- 3p **1d** Shown is a scatter diagram for educational levels (years of schooling completed) of husbands and wives in South Carolina, from the March 2005 Current Population Survey.



In this image, three areas are shaded. Which of the following descriptions is not indicated as a shaded area?

- (a) Husband completed 12 years of schooling and wife completed fewer years of schooling than husband.
 - (b) Husband completed more than 16 years of schooling.
 - (c) Wife completed 16 years of schooling.
 - (d) Wife completed more years of schooling than husband.
- 2p **1e** Which of the following options is **not** a real way that you could try collecting data:

- (a) Prospective Cohort Study
- (b) Lurking Factor Experiment
- (c) Randomized Controlled Experiment
- (d) Observational Study

- 4p **1f** A box contains red and blue marbles; there are 5 more red marbles than blue ones. Marbles are drawn one at a time from the box, at random with replacement. You win a dollar if a red marble is drawn more often than a blue one. There are two choices:
1. 100 draws are made from the box
 2. 200 draws are made from the box

Which of these options gives a better chance of winning?

Hint: Confidence limits.

- ☐ a Choice 2 gives a better chance of winning
 - ☐ b Can't tell without more information
 - ☐ c Choice 1 gives a better chance of winning
 - ☐ d Choices 1 and 2 give the same chance of winning
- 2p **1g** The central limit theorem (CLT) tells us something about distribution of?
Assuming that the requirements for CLT to apply hold.
- ☐ a Raw data
 - ☐ b Population parameters
 - ☐ c Sample means
 - ☐ d Root mean squared error
- 2p **1h** We run a machine learning algorithm to do classification to figure out when a house is in Los Angeles or in New York based on several features (size, price and so on). Our classification gives us an area under the ROC curve of 0.5. This means what?
- ☐ a High sensitivity
 - ☐ b Chance level of classifiability.
 - ☐ c Accuracy is slightly better than chance.
 - ☐ d 50% specificity.
 - ☐ e None of the above.

- 2p 1i Time freshmen spend at parties: A newspaper article says that on the average, college freshmen spend 7.5 hours a week going to parties. One administrator does not believe that these figures can be this high at her college, which has nearly 3,000 freshmen. She takes a simple random sample of 100 freshmen, and interviews them. On average they report 6.6 hours a week going to parties, and the SD is 9 hours.
- Which of the following hypotheses align with this example:
- (a) H_0 : 7.5 hours a week are spent going to parties AND H_a : more than 7.5 hours a week are spend going to parties
 - (b) H_0 : 6.6 hours a week are spent going to parties AND H_a : less than 6.6 hours a week are spend going to parties
 - (c) H_0 : 7.5 hours a week are spent going to parties AND H_a : less than 7.5 hours a week are spend going to parties
 - (d) H_0 : 6.6 hours a week are spent going to parties AND H_a : more than 6.6 hours a week are spend going to parties
- 2p 1j What visualization would you definitely not want to use to represent baby birth weight data? This dataset comprises of millions of data points representing the birth weights of babies collected over a year.
- (a) Box plot
 - (b) Line plot
 - (c) Strip chart
 - (d) Histogram
- 2p 1k A researcher records each participant's gender, age, and number of times they visited a doctor in the past year. Which pairing below **correctly** classifies the types of these three variables in order?
- (a) (Categorical, Continuous, Continuous)
 - (b) (Continuous, Categorical, Discrete)
 - (c) (Categorical, Continuous, Discrete)
 - (d) (Continuous, Continuous, Categorical)

- 3p **11** We are given the following probability distribution function:

$$f(x) = \frac{(7-2x)}{2} \text{ for } 2 \leq x \leq 3 \text{ and } f(x) = 0 \text{ otherwise.}$$

What is the population mean for this distribution?

- (a) 2.112 (b) 2.417 (c) 3.419 (d) 1.551

Judging statistical claims

The news article below claims that cat parasites cause brain cancer.

In the article they state: "Across the range of infection prevalence, from 4% to 67% of the population, *T. gondii* [cat parasite] was associated with a 1.6-fold increase in the risk of brain cancer."

UK NEWS WEBSITE OF THE YEAR 2024

The Telegraph

News Sport Business Money Opinion Ukraine Travel Health Lifestyle Culture Puzzles Podcasts

Cat parasite linked to brain cancer

A parasite spread by cats could almost double their owner's chance of developing brain cancer, research suggests.

"Cat parasite linked to brain cancer"

- 4p **2a** Assuming that 2 in 100 people gets brain cancer in low cat parasite prevalence countries. What is the expected frequency of getting brain cancer in countries with high prevalence of the cat parasite? Assume the 1.6 was an odds ratio and that we want to know the rates per 100 people.

- (a) None of the above
(b) 3.9/100
(c) 3.6/100
(d) 5/100
(e) 3.2/100

- 2p **2b** In reality, it turns out that cats do not cause cancer. A follow-up study that prospectively analyzed women with cats found confirmed that the cats cause cancer story was wrong. They created the following contingency table:

	Own Cat	Don't Own Cat
Brain Cancer	171	645
No Brain Cancer	114614	378066

What is the relative risk of getting brain cancer when you have a cat compared to when you do not have a cat?

- a

Conditional Probability

Before going on vacation for a week, you ask your spacey friend to water your ailing plant. Without water, the plant has a 70 percent chance of dying. Even with proper watering, it has a 10 percent chance of dying. And the probability that your friend will forget to water it is 25 percent.

- 9p

- (a) What's the chance that your plant will survive the week? [3 points]

- (b) If it's dead when you return, what's the chance that your friend forgot to water it? [3 points]

- (c) If your friend forgot to water it, what's the chance it'll be alive when you return? [3 points]

Design a Research Study

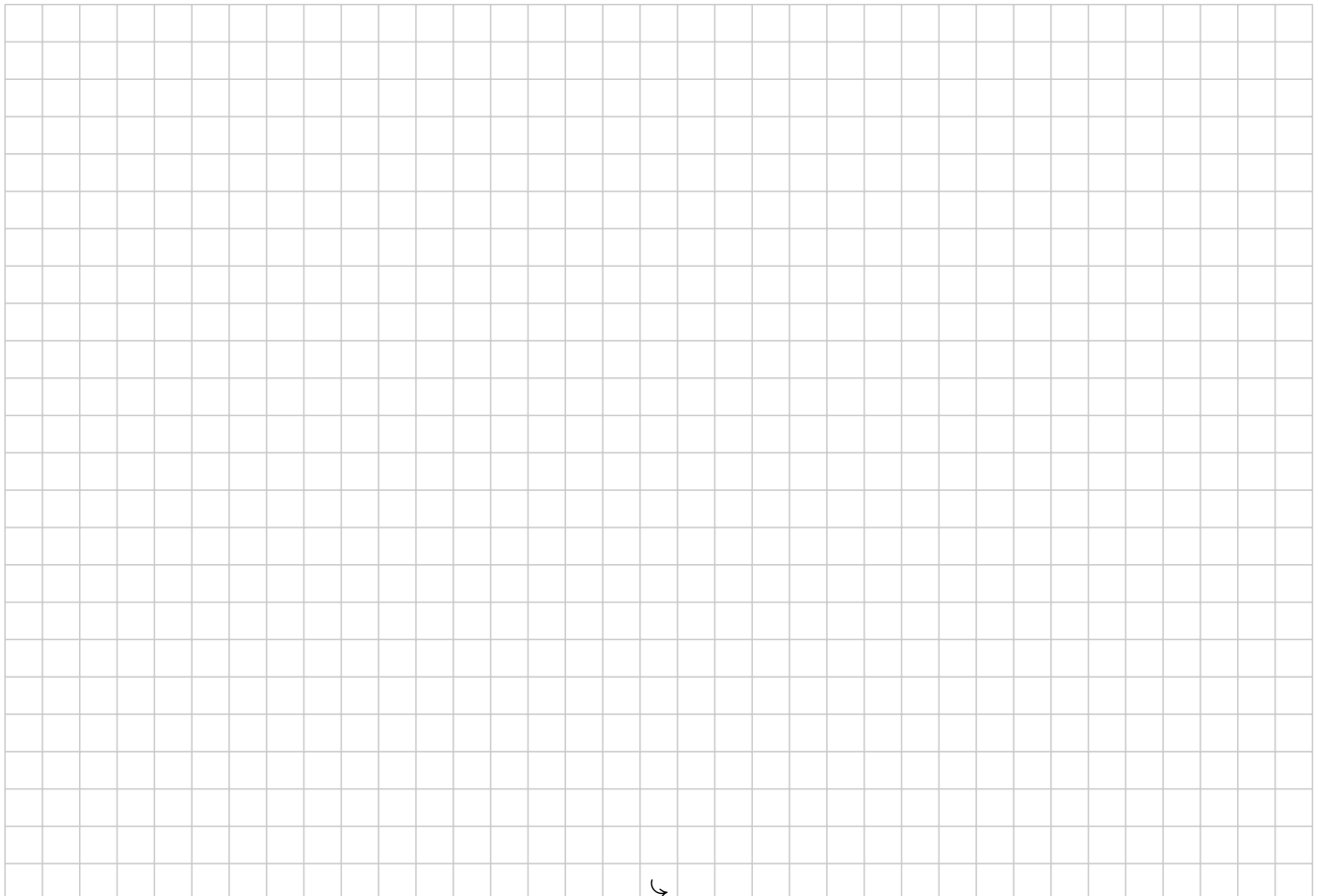
A regional government has issued a large study of the high schools in the region. As part of this study, an overview has been made of both the central exam and the school exam results of all students in the region. Note that in the Netherlands, someone's graduation grades are determined based on both these results.

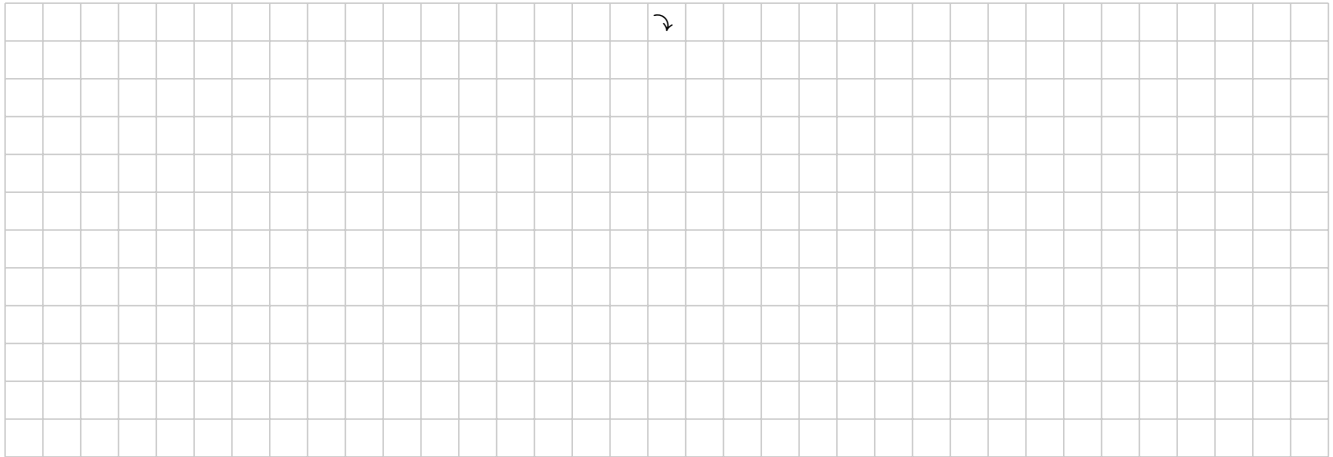
For each student, it is listed what grades were obtained for which course. Moreover, it is also known which teachers taught each course to the students and at which school they taught the course. Based on the average grades per school (for the central exam) that were calculated based on this information and subsequently communicated to the schools, the top school (highest average grades on the central exam) requests another analysis to be performed on the data.

8p **4 The school requests the following analysis to be performed:**

1. How do our students perform on each subject relative to the other schools in the region? Specifically, are there courses where our students under-perform relative to the other schools in the region?
2. Are the findings about performance across courses consistent across teachers?

Describe your approach to process the data described above (note that it has already been collected) in such a way as to answer the research questions above. Consider the full process of setting up the analysis. Do not worry about how this would be done in Python/elsewhere (that is, don't write pseudocode, focus on how the analysis would be performed and what you would assess).





Video game genres

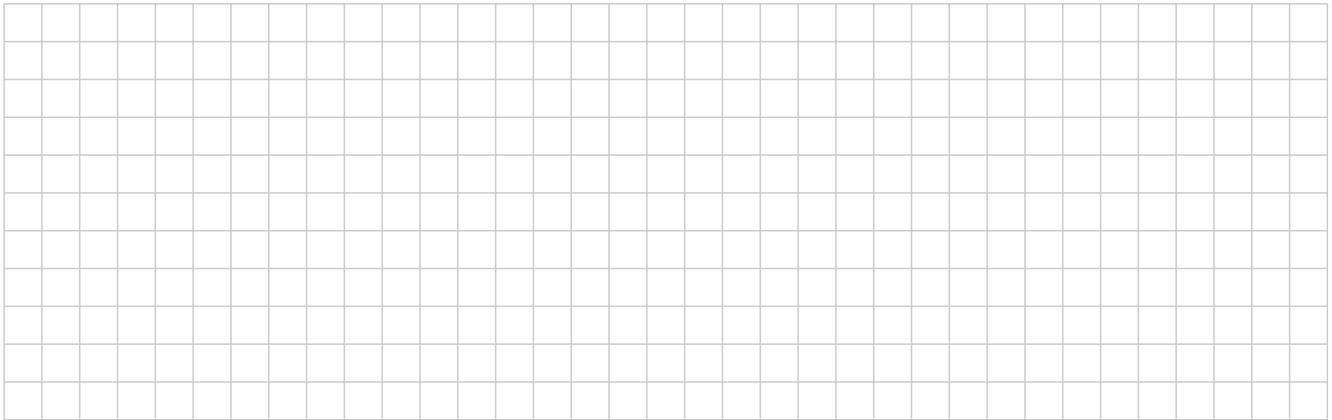
A survey of 1200 teens collected data about video game use by teens. According to the survey, the following are among the most popular game genres:

Genre	Examples	Percent who play
Racing	NASCAR, Mario Kart, Burnout	70
Sports	Madden, FIFA, Tony Hawk	65
Rhythm	Guitar Hero, Dance Dance Revolution, Lumines	61

- 8p **5a** Give a 95% confidence interval for the proportion who play games in each of these three genres. Make sure to note the correct formula and how/why you are applying it to calculate the confidence limits.



- 4p **5b** Explain why the chance that at least one of your intervals does not contain the true value of the parameter (that is, the population percentage of teens playing any of these genres of games) is greater than the α of 5% that we set for the confidence interval in part (a).



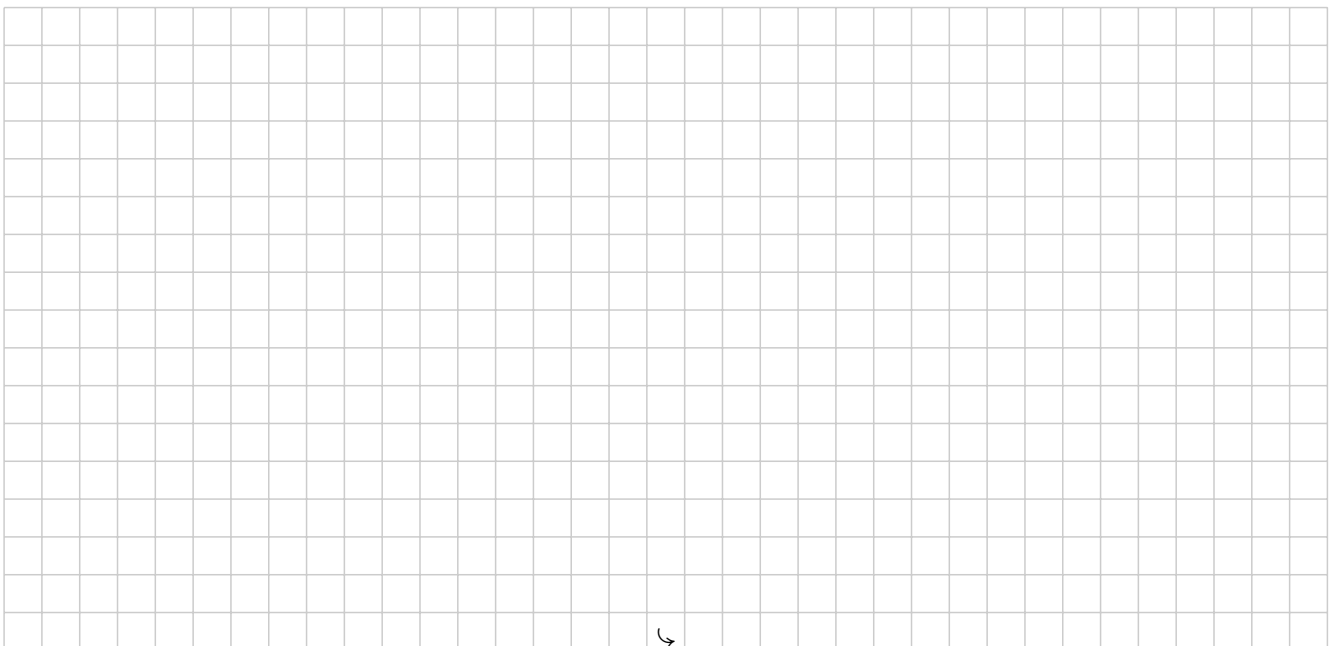
Box of cereal

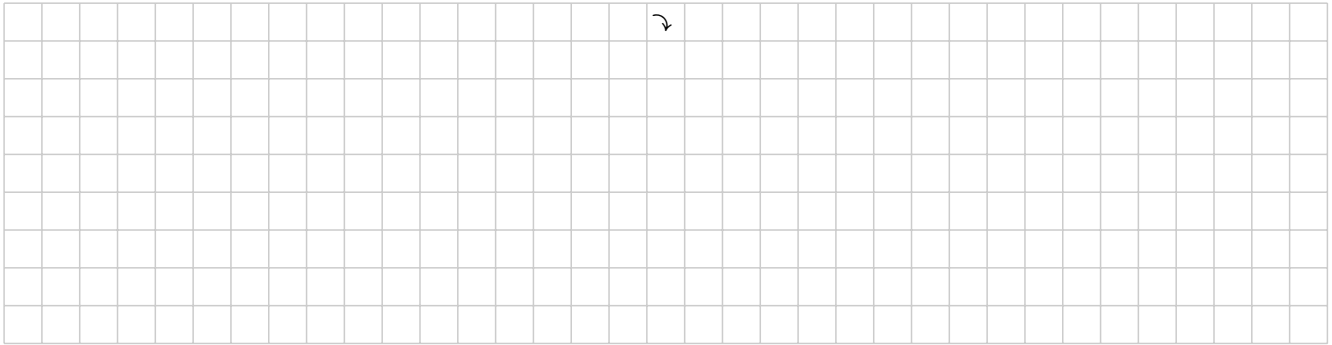
A packaging company is asked by a producer of cereal to fill boxes of their newest brand. The cereal company would like to have at least 200 grams of cereal in their boxes and they are willing to accept that in general, five out of a thousand boxes are filled with less than 200 grams.

Suppose the packaging company has a filling machine that fills boxes with a weight that is normally distributed with an unknown mean and a known standard deviation of 4 grams.

Upon an inspection of 16 boxes that were filled by the machine, the company found that the boxes were filled with sample mean 211 grams.

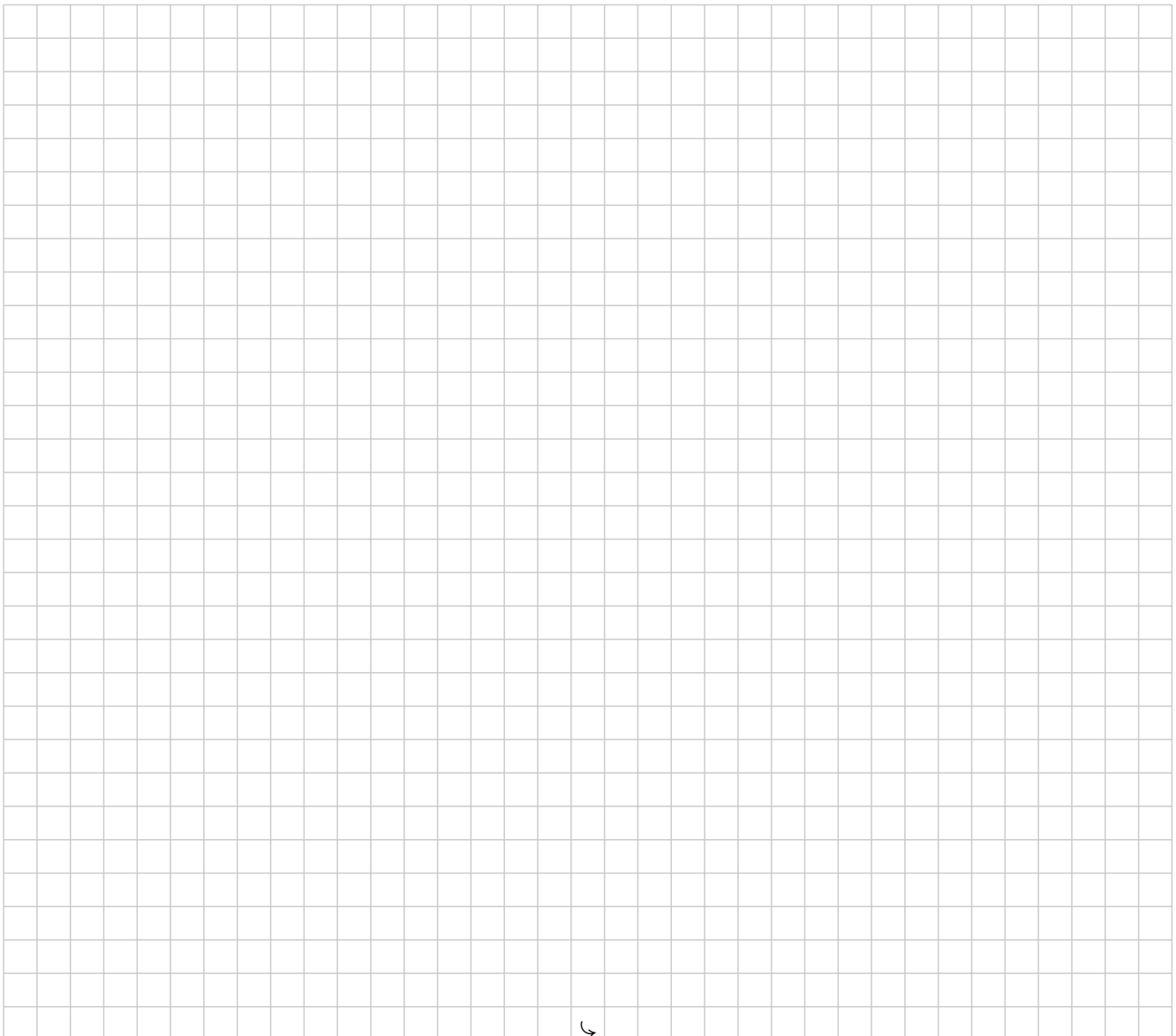
- 4p **6a** Construct a 95% confidence interval of the mean for the weight of the cereal box.





After further analysis of the initial output of the machine, the company finds that the machine fills boxes with a mean of 211 grams.

- 8p **6b** Based on this information, does the machine adhere to the standards that the cereal company demanded (i.e., at most five out of a thousand boxes are filled with less than 400 grams)?



Measurements are the core of data collection. Measurements are all around us in different forms: how many times did you visit Instagram today? how many coffees did you drink? how much money did you spend on your food? But measurements come with many caveats. Answer the following questions about the potential caveats of measurements:

- [Multiple correct answers possible, but negative points for incorrect answers are possible too.]

- ☐ A measure becoming the target.
- ☐ Measurements can be wrong.
- ☐ "Teaching to the test"
- ☐ Governments are not very clever.

- [Multiple answers possible, but negative points are possible too.]

- ☐ The meaning of the measurements and the values themselves may change as a function of where and when I make the measurements.
- ☐ The measurements we generate from the definitions of variables may not represent the real world.
- ☐ The measurements we get from the variables we define do not allow us to abstract the real world into numbers.
- ☐ We can be forced to create thresholds in defining the variables that affect the measurement.
- ☐ The measurements based on the variable definitions have too much spread.

Suppose the correlation coefficient between an independent and a dependent variable is -0.80 and the spearman rank correlation is -0.75 .

Statement: Below-mean values of the dependent variable (relative to its own mean) are associated with below-mean values of the independent variable (relative to its own mean).

2p **8a** Is the statement true or false?

- (a) True (b) False

4p **8b** Your motivation on why the statement is either true or false and why the spearman rank correlation matters to explain this:

RCTs and Causality

2p **9a** When analyzing two populations of fish, we want to understand whether changes in the population of one group of fish affects another population of fish. A first basic analysis tries to look at the correlation between the populations and finds that the correlation is close to 0. We can interpret this to mean:

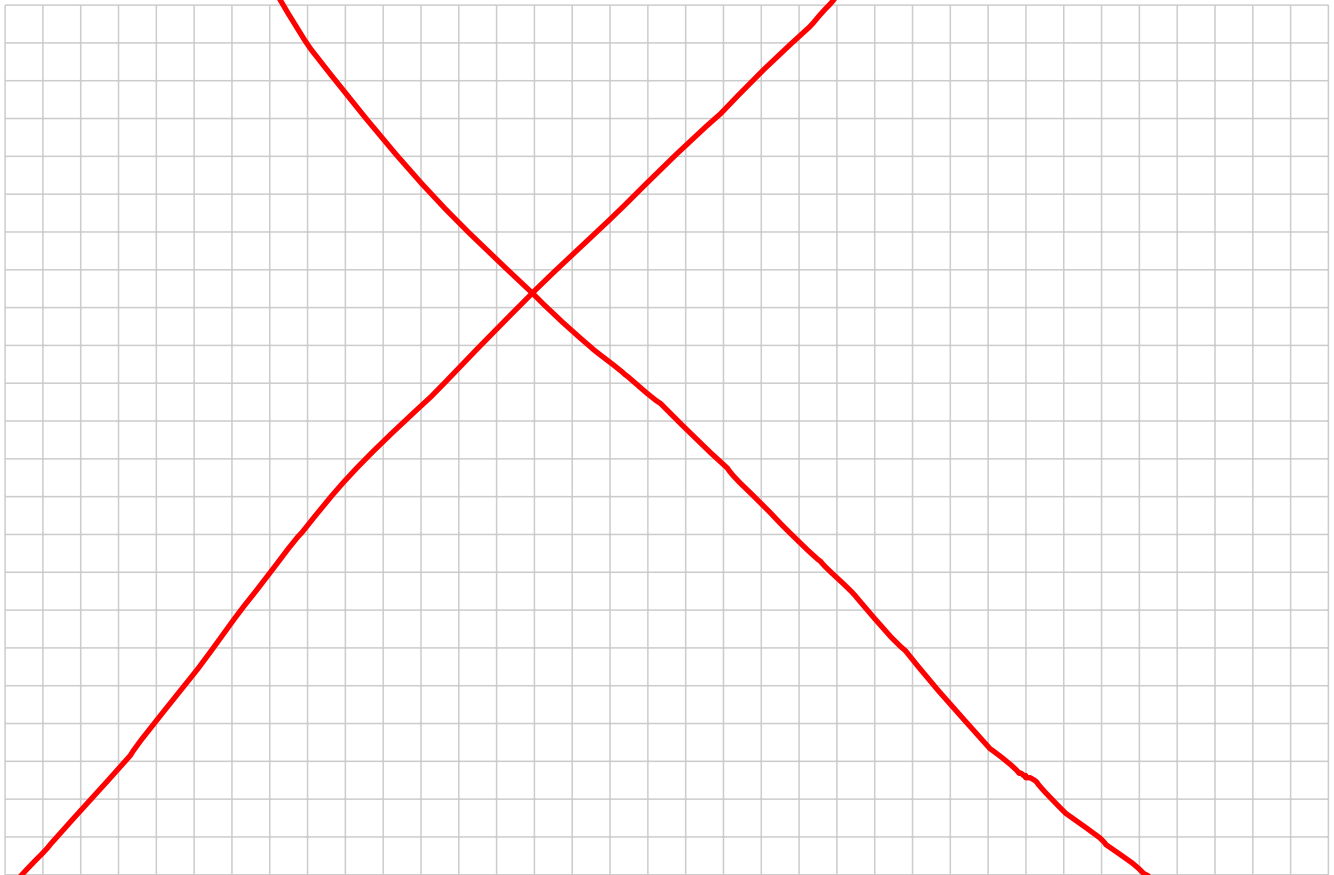
- ☐ a We can say nothing about causality from this result.
- ☐ b There is no causal relationship between the fish populations.
- ☐ c None of the above
- ☐ d There may still be a causal relationship between fish populations.

Pseudo-Coding Statistics

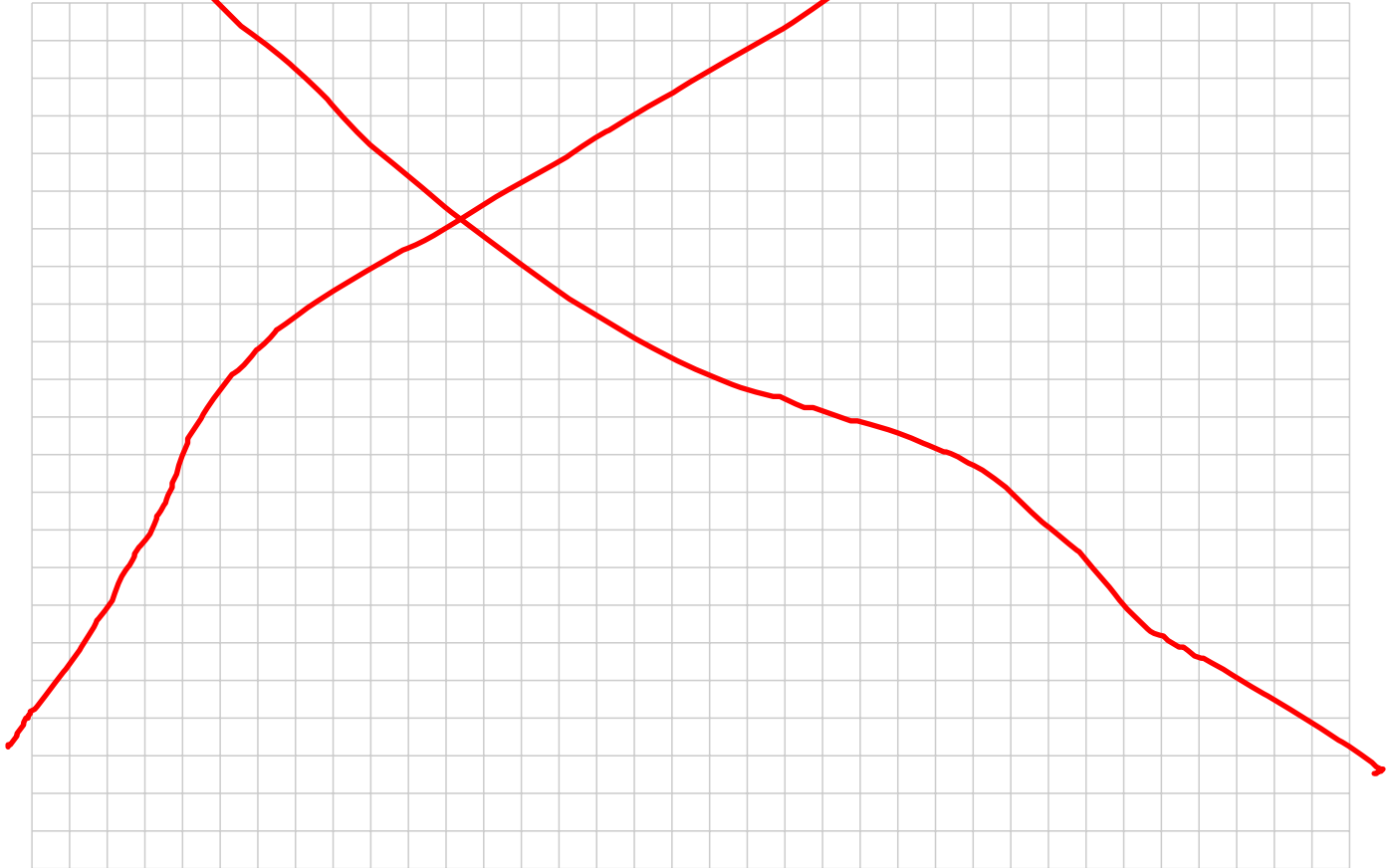
- 3p **10a** You are given a dataset consisting of incomes from men and women from all the states of the USA as well as some demographic information including family size, race, house ownership, gun ownership, house size, religiosity and political leaning. Unfortunately, each state gives you data in a separate comma-separated values (CSV) file and labels missing data differently, but always with a string. Every row of the dataset from each state represents a different individual and each column represents each of the characteristics already noted, including one column for income.

Write pseudo-code that allows you to answer the following questions:

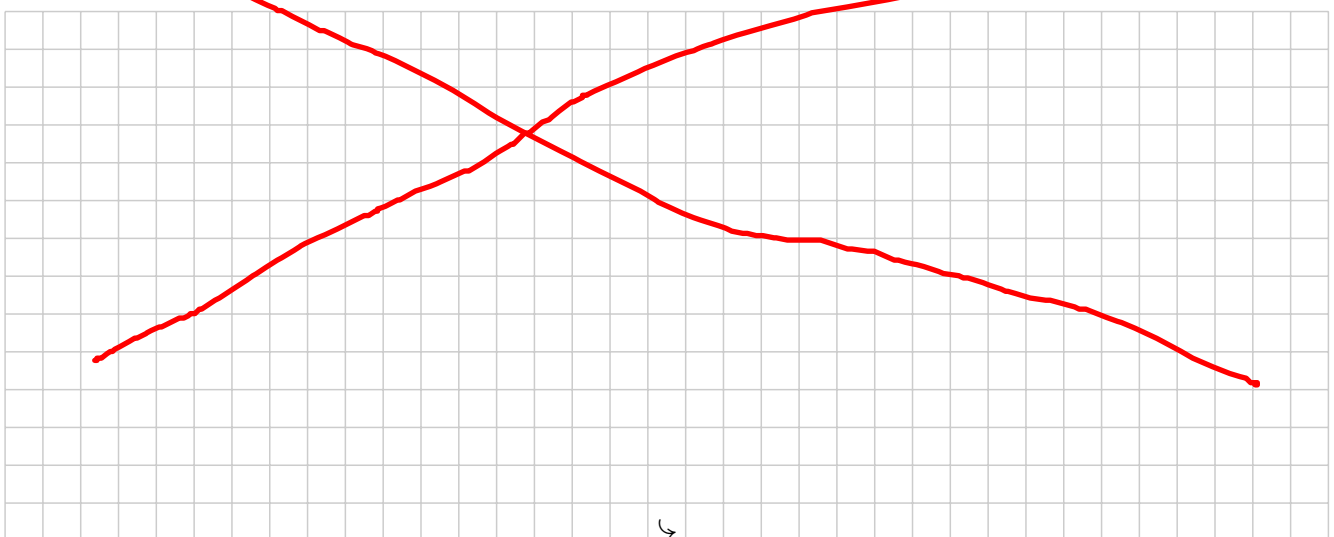
Given a dataframe with missing values, do the preprocessing (cleaning the missing data), and merge into a single data frame.

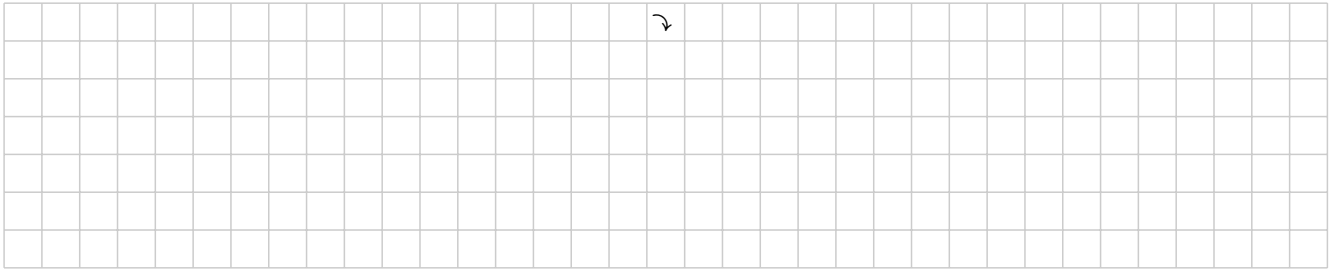


- 4p **10b** Assuming you already have a clean, pre-processed dataframe (collapsed across states and missing values are accounted for), write pseudo-code to assess whether there is any difference in incomes between men and women across the USA and specifically, within different states of the USA.

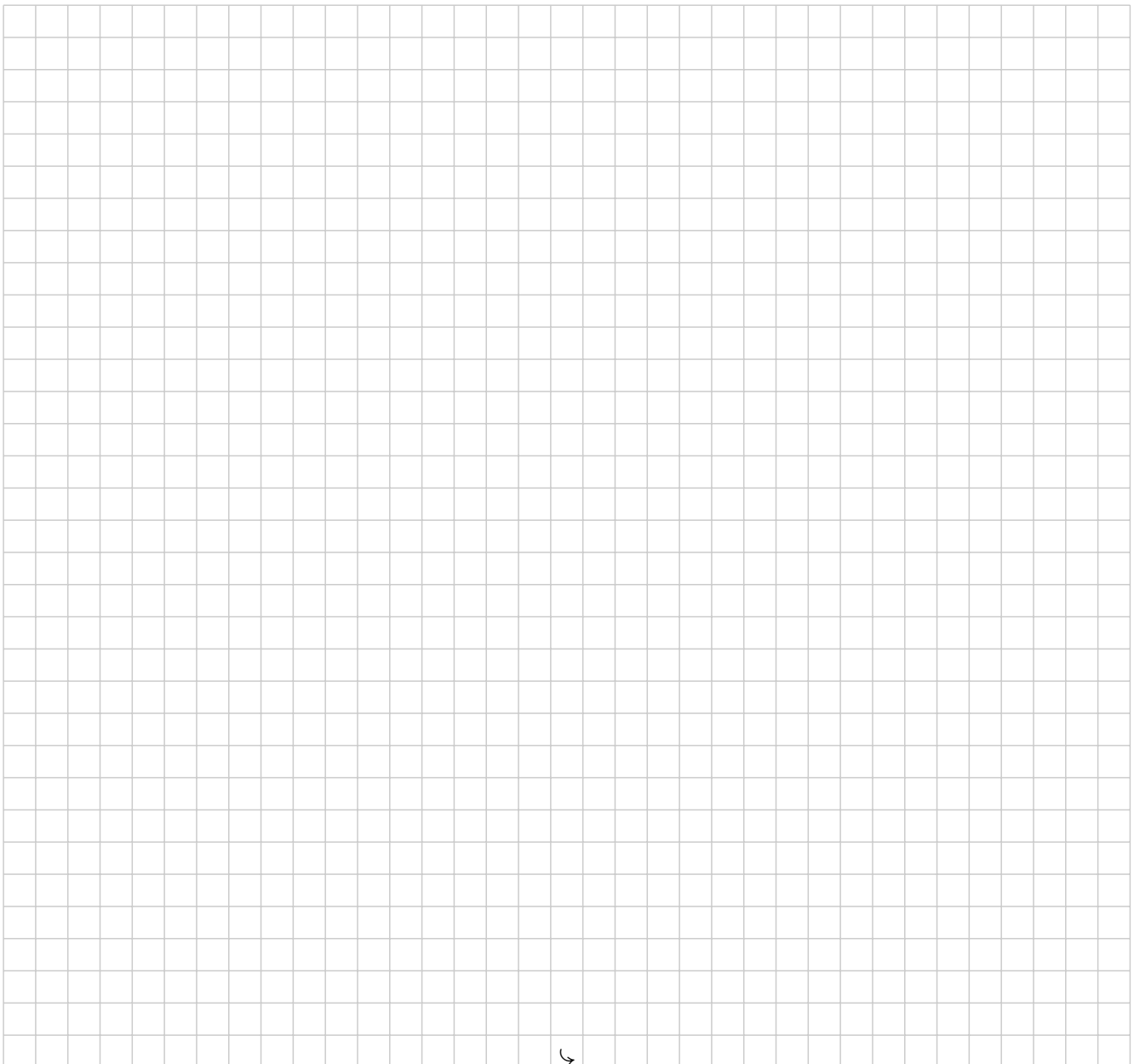


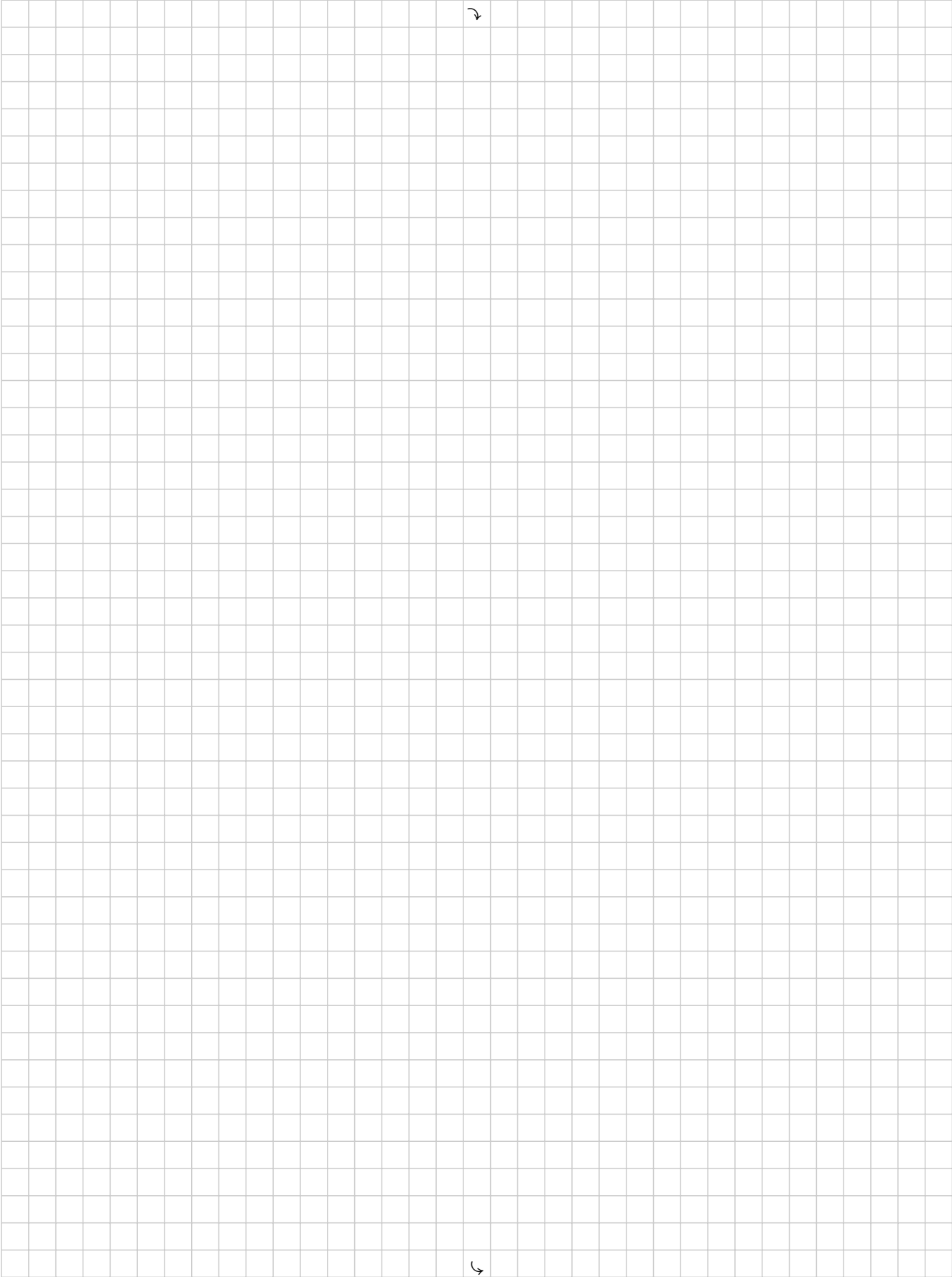
- 3p **10c** Assuming you already have a clean, pre-processed dataframe (collapsed across states and missing values are accounted for), write pseudo-code to classify which state any individual person is from based on the available information - make a note of how you would train this model and test it.

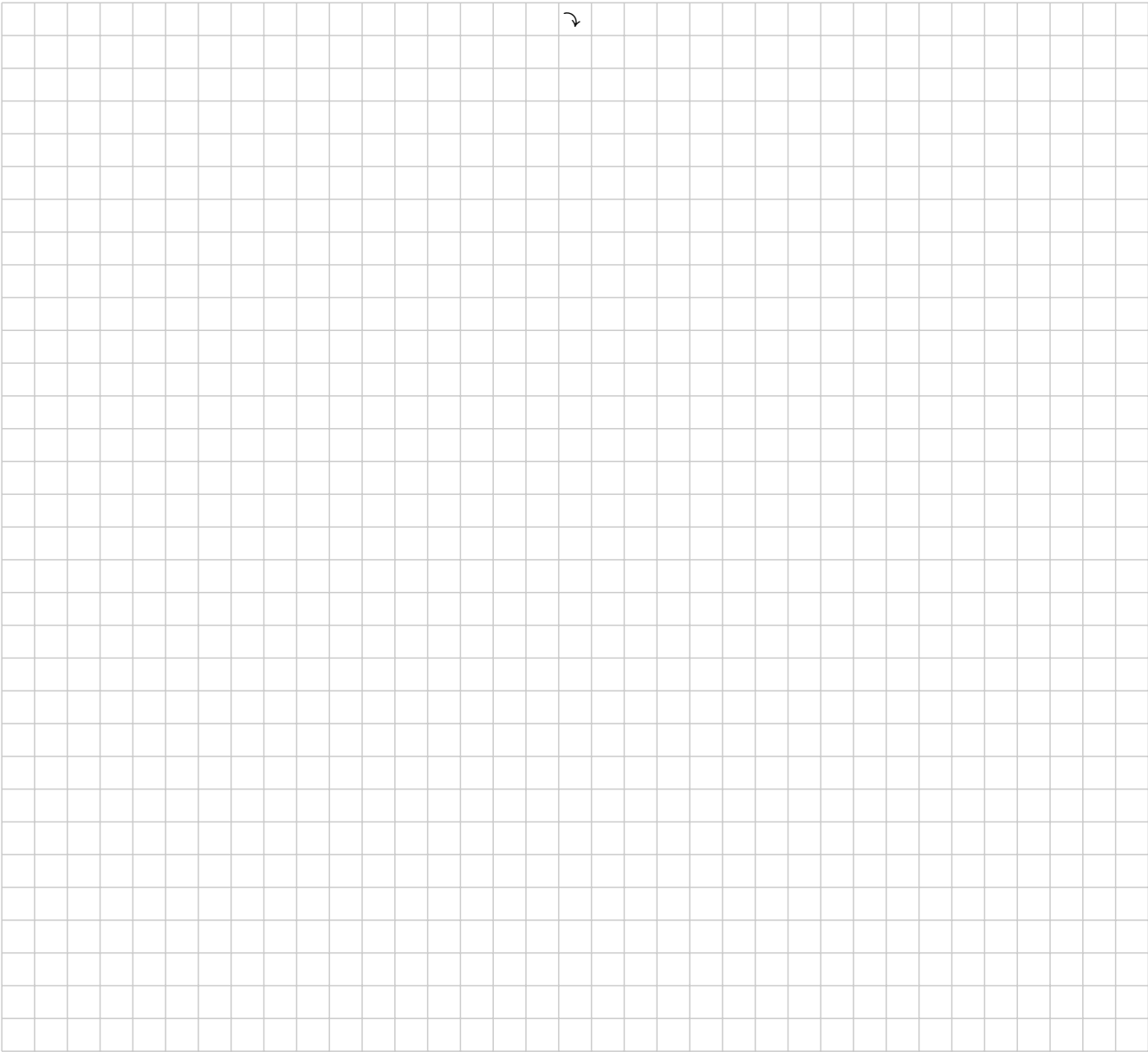


**Extra Space**

- 11** Use the space below if you ran out of space for other questions. Clearly indicate which question you are answering.







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